

Online Food Delivery System

Case Study: Online Food Delivery System (Multithreading Simulation)

1. Introduction

With the rapid growth of online services, food delivery platforms have become an essential part of daily life. Applications like Swiggy and Zomato handle multiple users, restaurants, and delivery agents simultaneously.

This project simulates such a real-world system using **multithreading in Java**, where different activities occur concurrently to improve efficiency and responsiveness.

2. Objective

The objective of this project is to:

- Simulate **multiple customers placing orders simultaneously**
- Enable **restaurants to process orders concurrently**
- Allow **delivery agents to deliver orders independently**
- Demonstrate **thread synchronization and communication**

3. Problem Statement

In a real-world food delivery system:

- Customers place orders at unpredictable times
- Restaurants prepare orders based on availability
- Delivery agents pick up and deliver orders

If not handled properly, this can lead to:

- Race conditions
- Data inconsistency
- Order mismanagement

Hence, a **thread-safe system** is required.

4. System Design

4.1 Components

Component	Description
Customer Thread	Generates and places orders
Restaurant Thread	Prepares orders
Delivery Agent Thread	Delivers prepared orders
Shared Queue	Stores orders safely

5. Workflow

Customer → Places Order → Order Queue

Restaurant → Takes Order → Prepares → Ready Queue

Delivery Agent → Picks Order → Delivers

6. System Architecture

[Customer Threads]



[Order Queue] ← synchronized



[Restaurant Threads]



[Ready Queue]



[Delivery Threads]

7. Output Explanation

- Customers place orders concurrently
- Restaurant prepares orders one by one
- Delivery agents deliver orders independently
- Output shows interleaving of thread execution